Reconstruction-Assisted and Distance-Optimized Adversarial Training: A Defense Framework for Remote Sensing Scene Classification

Yuru Su, Ge Zhang, *Graduate Student Member, IEEE*, Shaohui Mei, *Senior Member, IEEE*, Jiawei Lian, *Graduate Student Member, IEEE*, Ye Wang, *Graduate Student Member, IEEE*, and Shuai Wan, *Member, IEEE*

***Abstract*— Despite deep neural networks (DNNs) have been widely applied in remote sensing (RS) scene classification and achieved satisfying performance, the vulnerability of DNNs toward adversarial examples significantly degrades their performance. Moreover, the relatively limited labeled samples of RS scene classification make DNNs more likely to overfit, leading to weak generalizability and noise sensitivity. This may result in DNNs being more vulnerable to adversarial examples. Consequently, the defense of adversarial examples is of crucial importance to improve both the generalizability and robustness of DNNs in the RS scene classification task. However, few studies have been conducted on defense for RS scene classification, especially ignoring the intrinsic characteristics of RS images. In this article, an effective defense framework for RS scene classification, named reconstruction-assisted and distance-optimized adversarial training (RDAT), is proposed to defend adversarial examples. To solve the problems caused by high interclass similarity, a distance-optimized (DO) strategy is designed for adversarial training (AT) to strengthen the learning of underfitting content, increase the interclass distance, and improve the robustness of the networks. Furthermore, to generate high-quality samples for AT, a reconstruction-assisted (RA) block is proposed to eliminate adversarial perturbations in adversarial examples. Specifically, in this block, by Swin Transformer (SwinT) block and multiscale convolution (MSC) block, SwinT-MSC-UNet (SMUNet) is constructed to fully extract global and multiscale local features to adapt to the characteristics of RS images with a large variance of ground object scales. Extensive experiments on the benchmark datasets, that is, UC Merced (UCM) and aerial image dataset (AID), have demonstrated that the proposed RDAT can effectively resist multiple adversarial attacks and yield superior results than other defense methods for RS scene classification.**

***Index Terms*— Adversarial defense, adversarial training (AT), image reconstruction, remote sensing (RS), scene classification.**

I. Introduction

With the remarkable development of remote sensing (RS) technology, numerous high-resolution RS images are available [1], [2], [3], [4]. RS scene classification refers to classifying RS images into predefined categories based on their content, which has been widely applied in land cover classification [5], natural hazard detection [6], urban planning [7], and so on. For this task, the main challenges include big intraclass diversity, high interclass similarity, large variance of ground object scales, and difficulty in extracting features with discriminative properties [8].

In the past decades, plenty of methods have been proposed for RS scene classification. The traditional methods [9], [10], [11], [12], [13] are handcrafted feature-based and greatly depend on the professional knowledge, which are difficult to meet the needs of increasingly complex RS image scenes due to the limited representation capacity [14], [15]. Deep neural networks (DNNs) have the advantage of end-to-end feature learning to automatically extract abstract high-level features. DNNs have found extensive applications in the RS field [16], encompassing tasks such as hyperspectral image processing [17], [18], [19], synthetic aperture radar (SAR) image analysis [20], [21], [22], and optical RS processing [23], [24]. Among these applications, numerous DNN architectures have demonstrated outstanding classification performance, including GoogLeNet [25], ResNet [26], DenseNet [27], ResNeXt [28], and EfficientNet [29].

Despite DNNs having achieved great performance on RS scene classification, they are still vulnerable to adversarial examples. The adversarial examples are those containing elaborate perturbations invisible to humans. Due to DNNs' nonintuitive traits and inherent blind spots, it is easy to misclassify while facing an adversarial example [30], [31], [32]. Meanwhile, since the labeled samples for RS scene classification are relatively limited [33], [34], DNNs are easier to overfit, resulting in poor generalization and noise sensitivity. In this case, DNNs may be more vulnerable to adversarial examples [35]. As shown in Fig. 1, with the addition of perturbations, DNNs output misclassification with high confidence, limiting the use in safety-critical areas such as military intelligence and security, environmental monitoring, and RS image transmission security. Therefore, it is necessary to study the adversarial defense for RS scene classification.

To improve the robustness of the networks, a large number of defense approaches have been proposed. Goodfellow et al.

Manuscript received 7 June 2023; revised 7 October 2023; accepted 17 October 2023. Date of publication 31 October 2023; date of current version 9 November 2023. This work was supported by the National Natural Science Foundation of China under Grant 62171381. *(Corresponding author:Shaohui Mei.)*

The authors are with the School of Electronics and Information, Northwestern Polytechnical University, Xi'an 710129, China (e-mail: suyuru_nwpu@mail.nwpu.edu.cn; zhangg@mail.nwpu.edu.cn; meish@nwpu.edu.cn; lianjiawei@mail.nwpu.edu.cn; wy2017263322@mail.nwpu.edu.cn; swan@nwpu.edu.cn).

Digital Object Identifier 10.1109/TGRS.2023.3328889

1558-0644 © 2023 IEEE. Personal use is permitted, but republication/redistribution requires IEEE permission.
See https://www.ieee.org/publications/rights/index.html for more information.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 19:39:23 UTC from IEEE Xplore. Restrictions apply.

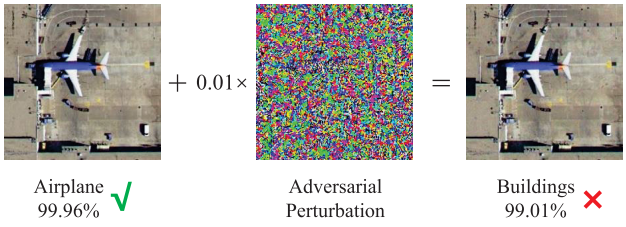


Fig. 1. Process of generating an adversarial example. The left clean image is correctly classified as "Airplane" with 99.96% probability; however, with the elaborate perturbation shown in the middle image, although humans cannot recognize the change, the image is classified in the wrong category "Buildings" with 99.01% probability.

[30] first proposed the concept of adversarial training (AT) to improve the robustness of the networks by training with adversarial examples. Madry et al. [36] theoretically investigated AT and gave a min-max formulation from the perspective of robust optimization. Papernot et al. [37] introduced a defensive distillation method to make the gradient smoother and reduce the sensitivity of the networks to perturbations. Samangouei et al. [38] proposed a Defense-GAN, which uses the trained generator to map the input adversarial examples to the distribution of clean examples to mitigate the effects of adversarial perturbations. Zhang et al. [39] identified a tradeoff between robustness and accuracy, demonstrating an upper bound on the gap between robust error and optimal natural error, and then constructed a defense method TRADES. However, these methods mainly focus on the field of nature images (e.g., CIFAR-10 dataset) with insufficient adaptation to RS images. In addition, Chen et al. [40] proposed a soft threshold defense method for RS images based on which it will be rejected if the input image is detected as an adversarial example. Cheng et al. [41] developed a perturbation-seeking generative adversarial network (PSGAN) for RS scene classification, which models the distribution of adversarial perturbations and is effective against unknown attacks. It is worth noting that RS images present intrinsic characteristics: high interclass similarity and large variance of ground object scales [42]. The presence of identical objects within different scene categories or the substantial semantic overlap among scene classes result in a high degree of interclass similarity. This situation can easily lead the networks to extract high-dimensional features with close distances on different classes, affecting the robustness of the networks. In addition, the diversity in shooting distances and observational conditions in RS images results in noticeable scale variations within the same category of objects. This poses particular challenges for scene classification algorithms and their robustness studies. However, the image reconstruction employed in PSGAN is generic and lacks specific enhancements tailored to the scale characteristics of RS images. Furthermore, the training process of the classification networks does not consider the high interclass similarity often present in RS images, resulting in limited adaptability to RS datasets.

To address these issues, in this article, a reconstruction-assisted and distance-optimized AT (RDAT) defense framework is proposed for RS scene classification. In the proposed framework, the distance-optimized (DO) strategy is introduced with the aim of increasing high-dimensional interclass distances to address the issue of high interclass similarity. The DO strategy-based AT consists of three parts: base AT, intensive training, and restorative training. It focuses on identifying categories with close feature distances and enhancing the network's learning of the underfitting content to improve the defense ability of the network against adversarial examples. Besides, a reconstruction-assisted (RA) block is adopted to mitigate the effects of the adversarial perturbations with SwinT-MSC-UNet (SMUNet). Within SMUNet, the Swin Transformer (SwinT) block focuses on exploring long-range dependencies, enabling the extraction of global information and a comprehensive understanding of the scene. Meanwhile, the multiscale convolution (MSC) block aggregates convolutional kernels with different receptive fields, allowing the capture of multiscale local features and enhancing the understanding of image details. Moreover, the loss function of SMUNet includes two components, visual similarity and feature similarity, which constrain the reconstructed image to be similar to the clean example not only in visual space, but also in feature space. Finally, clean examples, adversarial examples, and reconstructed examples are mixed for DO strategy-based AT to build a robust network.

Extensive experiments are conducted on two widely used benchmarks: UC Merced (UCM) and aerial image dataset (AID). The results show that the proposed defense framework RDAT achieves good performance on three target networks, that is, ResNet152 [26], ResNeXt101 [28], and SCNet [43], without changing their structure. The main contributions of this article are summarized as follows.

1) To address the adversarial defense challenges in RS scene classification and improve the robustness of the network toward adversarial examples, an effective defense framework named RDAT is proposed, in which both high interclass similarity and large variance of ground object scales are fully considered, adapting to the characteristics of RS images.
2) To address the impact of high interclass similarity and enhance the robustness of DNNs, a DO strategy-based AT is proposed. This strategy aims to increase the high-dimensional feature distance between classes, thus strengthening the adaptability of networks to adversarial perturbations.
3) To effectively mitigate adversarial perturbations and accommodate the large variance of ground object scales in RS images, in the RA block, SMUNet is constructed to generate reconstructed examples, which can extract rich discriminative features through a combination of the global modeling capability of SwinT and the multiscale local modeling capability of MSC.

The rest of this article is arranged as follows. In Section II, the related works are briefly introduced. In Section III, the proposed RDAT is introduced in detail. Section IV presents the experimental settings and comprehensive results. Conclusions are finally summarized in Section V.

II. Related Work

A. Adversarial Attack

Szegedy et al. [44] first demonstrated the concept of adversarial examples. After adding subtle perturbations

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 19:39:23 UTC from IEEE Xplore. Restrictions apply.

imperceptible to the human eyes to a correctly classified image, the networks output an incorrect classification with a very high confidence level, where the image with perturbation is called an adversarial example. The method of attacking DNNs using adversarial examples is generally called an adversarial attack.

In the classification task, given a trained DNN as $f(\boldsymbol{x}) = y$, $\boldsymbol{x}$ is a clean image with the corresponding label y. There are two kinds of adversarial attacks: untargeted [31] and targeted [44]. For the input $\boldsymbol{x}$ and the network $f(\cdot)$, the untargeted attack is to search for a small perturbation r that makes $f(\boldsymbol{x}+r) \neq f(\boldsymbol{x}) = y$. By contrast, unlike untargeted attacks that are only misclassified, targeted attacks aim to obtain a specified label y', that is, $f(\boldsymbol{x}+r) = y'$. To make the perturbation small enough without being detected by humans, most attack algorithms use L_p distance between the clean image $\boldsymbol{x}$ and the adversarial image $\boldsymbol{x}+r$ to limit r, where $p \in \{1, 2, \infty\}$.

From another perspective, depending on the attacker's knowledge of the target network, adversarial attacks can be divided into white-box [45] attacks and black-box [46] attacks. In a white-box attack, the attacker can obtain all information about the target network $f(\cdot)$, including the structure and parameters of the network. In a black-box attack, the attacker has no access to the information about the target network except for its input and output. Obviously, white-box attacks are easier to conduct compared to black-box attacks. In terms of current research, white-box attacks are the main way to generate adversarial examples. Several white-box attacks used in this article are presented below.

1) Fast Gradient Sign Method: Goodfellow et al. [30] proposed a gradient-based single-step attack method, fast gradient sign method (FGSM), which seeks the direction with the most significant change in network gradient and change the corresponding pixel values only once for every image.

Given a clean image $\boldsymbol{x}$ with the corresponding label y, the adversarial perturbation r, and the adversarial image $\boldsymbol{x}_{\text{adv}}$ generated by FGSM can be expressed as

$$\begin{cases} r = \varepsilon \cdot \text{sign}(\nabla_{\boldsymbol{x}} J(\theta, \boldsymbol{x}, y)) \\ \boldsymbol{x}_{\text{adv}} = \boldsymbol{x} + r \end{cases} \tag{1}$$

where θ is the well-trained parameters of the target network and $\nabla(\cdot)$ is the gradient of the objective function $J(\cdot)$. Besides, $\text{sign}(\cdot)$ represents the sign function and ε controls the attack strength.

2) Project Gradient Descent (PGD): Madry et al. [36] proposed PGD, which is one of the benchmark testing algorithms, to evaluate the robustness of the networks. Different from FGSM, PGD employs multiple iterations and can be seen as an iterative version of FGSM. Each iteration clips the perturbation to a prescribed range, and this multistep iteration allows a better attack on the nonlinear network. The adversarial image $\boldsymbol{x}_{\text{adv}}^{t+1}$ can be calculated as

$$\boldsymbol{x}_{\text{adv}}^{t+1} = \prod_{\boldsymbol{x}+S} (\boldsymbol{x}^t + \alpha \cdot \text{sign}(\nabla_x J(\theta, \boldsymbol{x}, y)) \tag{2}$$

where α is the step size for each attack iteration and $\prod_{\boldsymbol{x}+S}(\cdot)$ limits the range of perturbations in $\boldsymbol{x}+S$.

3) Carlini and Wagner (CW) Attack: CW [47] developed a stronger optimization-based attack method, called CW. The algorithm regards the adversarial example as a variable with two constraints: one is the smaller, the distance between $\boldsymbol{x}$ and $\boldsymbol{x}_{\text{adv}}$ the better, and the other is the higher, the probability of misclassification the better. The optimization objectives of CW are as follows:

$$\begin{cases} \boldsymbol{r}_n = \dfrac{1}{2}(\tanh(\boldsymbol{\omega}_n) + 1) - \boldsymbol{x}_n \\ \min\limits_{\boldsymbol{\omega}_n} \ \|\boldsymbol{r}_n\| + c \cdot f\left(\dfrac{1}{2}(\tanh(\boldsymbol{\omega}_n) + 1)\right) \\ f(\boldsymbol{x}') = \max\left(\max\left\{Z(\boldsymbol{x}')_i : i \neq t\right\} - Z(\boldsymbol{x}')_t, -k\right) \end{cases} \tag{3}$$

where $\boldsymbol{r}_n$ is the distance metric between the clean example and the adversarial example and is optimized by $\boldsymbol{\omega}_n$. c is a hyperparameter to weigh the relationship between two constraints. $f(\cdot)$ is the loss function and $Z(\cdot)$ represents the output before softmax layer in the network. t is the target class wanted to be attacked into, and the optimization makes $\max\{Z(\boldsymbol{x}')_i : i \neq t\} - Z(\boldsymbol{x}')_t$ smaller, suppressing the logical value of the correct class while raising the target class t. The hyperparameter k denotes the confidence level related to the attack strength.

B. Adversarial Defense

With the continuous evolution of adversarial attack techniques, an increasing number of adversarial defense methods have emerged [48], [49]. The existing defense methods can be divided into four types: gradient masking, data preprocessing, adversarial example detection, and AT.

1) Gradient Masking: Inspired by knowledge distillation [50], Papernot et al. [37] proposed a defensive distillation method. First, the teacher network is trained to acquire prior knowledge at a distillation temperature of T. Then, the prior knowledge is fed to the student network at the same temperature for further training to obtain the final output. By adjusting the temperature during the distillation process, it is possible to yield a smoother network. Ross and Doshi-Velez [51] introduced input gradient regularization (IGR) to directly optimize the network, aiming to make the gradient of the network smoother so that small perturbations will not significantly affect the output. However, gradient masking methods are unsuitable for stronger attacks like CW.

2) Data Preprocessing: Data preprocessing aims to reduce the impact of adversarial perturbations by dealing with the adversarial examples before they are fed into the network. Dziugaite et al. [52] reduced the threat capability of the adversarial image by JPG compression to mitigate the impact of adversarial perturbations. However, it is determined that relying solely on compression is not an efficient means of defense. Based on the generative adversarial network (GAN), Samangouei et al. [38] proposed a Defense-GAN to remove perturbations by constructing an image close to the input example using the generative network, where the semantic information of the original image may be affected.

3) Adversarial Example Detection: Adversarial example detection is the process of detecting the presence of an adversarial example in the inference phase. Lu et al. [53] proposed

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 19:39:23 UTC from IEEE Xplore. Restrictions apply.

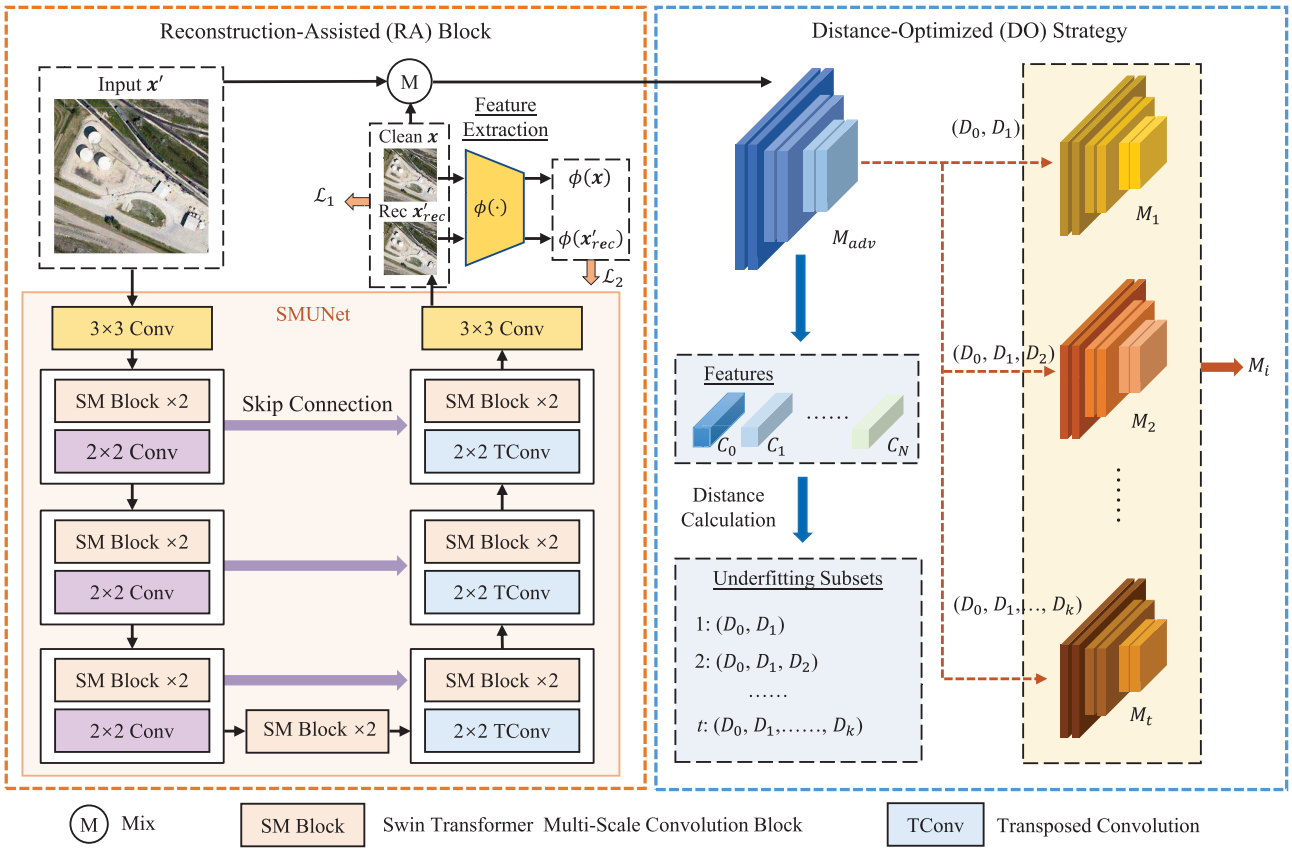


Fig. 2. Overview of the proposed defense framework RDAT. In the RA block, SMUNet transforms the adversarial examples into reconstructed examples, where the SM Block is utilized as the main building block. The training loss $\mathcal{L}$ of SMUNet consists of two parts: visual loss $\mathcal{L}_1$ and feature loss $\mathcal{L}_2$. Besides, "Conv" means convolution for downsampling; "TConv" means transposed convolution for upsampling. After image reconstruction, the DO strategy is executed to obtain robust networks. First, the high-dimensional features of each class are obtained from the network M_{adv} trained with base AT, then the feature distances are calculated to generate underfitting subsets (D_0, D_1), and so on. Finally, intensive training with underfitting subsets and restorative training with all data are performed in sequence to improve the robustness.

a SafetyNet, which consists of a classifier and an attack detector, and if the detector detects an adversarial example, the example will not be classified. Xu et al. [54] proposed a feature squeezing-based defense method. Comparing the prediction of the original input with the prediction of the compressed input, if it is larger than a certain threshold, the adversarial example is detected. However, when an attacker understands the detection principle, it can bypass the detection mechanism to attack successfully.

4) Adversarial Training: AT is generally considered by academics to be the most effective defense strategy against attacks so far. Madry et al. [36] first investigated it theoretically and formulated it through the perspective of robust optimization, which is a composition of an inner maximization problem and an outer minimization problem that can be expressed as follows:

$$\min_{\theta} \mathbb{E}_{(\boldsymbol{x},y)\sim\mathcal{D}}\left[\max_{\|r\|\leq\varepsilon} L(\theta, \boldsymbol{x}+r, y)\right] \tag{4}$$

where the adversarial perturbation r is restricted to the range ε, and θ is the parameter of the network. Inner maximization is to find the best perturbation r, generating adversarial examples to maximize the loss to attack the networks. Outer minimization is to learn the appropriate parameter θ to defend the adversarial example and improve the robustness of the network.

The defense framework RDAT combines data preprocessing and AT, where adversarial examples are reconstructed by SMUNet with global and multiscale local information, and a DO strategy-based AT is implemented to strengthen the learning of underfitting content to enhance the adversarial robustness of the network.

III. Proposed Method

A. Overview

Fig. 2 shows the proposed defense framework RDAT for RS scene classification. This framework mainly includes two parts, RA block and DO strategy-based AT. The RA block aims to eliminate the adversarial perturbation. It is realized by SMUNet, where the output reconstructed examples are forced to be as close as possible to the clean examples in both visual similarity and feature similarity. Besides, the DO strategy is proposed to improve defense capability and robustness from the perspective of distance, specifically, to make the network strengthen the learning of underfitting content. More details are discussed below.

B. RA Block

In the RA block, SMUNet is proposed to generate reconstructed examples. First, the encoder of SMUNet achieves

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 19:39:23 UTC from IEEE Xplore. Restrictions apply.

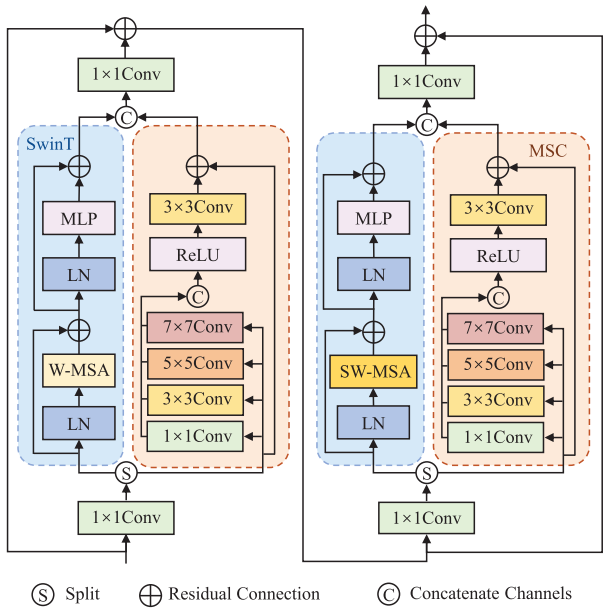


Fig. 3. SM block with two branches, SwinT and MSC. In SwinT, windows multihead self-attention (W-MSA) and shifted W-MSA (SW-MSA) alternate. MSC includes four scale convolution with kernels of 1×1, 3×3, 5×5, and 7×7, and the padding of the four scales are 0, 1, 2, and 3. The output feature maps of the SM block are the same size as the inputs, and they are summed together through the residual structure.

feature extraction and dimension conversion of the input examples through a 3×3 convolution with stride 1, then performs feature learning via the SM block and realizes downsampling by a 2×2 convolution with stride 2. Symmetrically, the decoder gradually recovers the image size by upsampling through a 2×2 transposed convolution with stride 2. Finally, a 3×3 convolution with stride 1 is used to recover the channel number to output the reconstructed image. Meanwhile, the encoder-decoder is connected by skip connection to retain more spatial information.

SM block is the main building block of SMUNet, where the input is divided into two branches in the channel dimension, passing through the SwinT block and the MSC, respectively, and finally concatenating together again. SwinT is introduced to extract global information by incorporating long-range dependencies. This assists the networks in gaining a better understanding of the overall scene, strengthening the restoration of global features. MSC captures rich multiscale local features by aggregating convolution kernels with different receptive fields, enabling a better understanding of the details of the image. In general, although the adversarial samples and clean samples are difficult to distinguish in the visual space, long-range dependencies, and local features still provide sufficient information in the feature space. The combination of global and multiscale local information enables the network to fully capture the feature patterns of adversarial and clean samples and make the networks adapt to the large variance of ground object scales of RS images, thus effectively eliminating adversarial perturbations before being recognized. The architectural details of the SM block are shown in Fig. 3.

In addition, for the optimization of the reconstructed network, two objective functions are adopted: visual similarity and feature similarity. For the first one, the goal is to make the reconstructed example as similar as possible to the original clean example from a human visual perspective; thus the L_1 loss is used to measure the distance between the reconstructed image and the clean image, as shown in the following equation:

$$\mathcal{L}_1 = \|T(\boldsymbol{x}') - \boldsymbol{x}\|_1 \tag{5}$$

where $\boldsymbol{x}'$ is the adversarial examples, generated by FGSM, PGD, or CW, and $T(\boldsymbol{x}')$ is the reconstructed examples generated by SMUNet.

Moreover, in addition to the reconstruction loss at the pixel level of human vision, the reconstruction loss at the feature level of networks is also necessary. To acquire feature similarity, a feature extraction module from the pretrained VGG16 network is added at the end of the SMUNet. This structure can be analogous to the discriminator of the GAN, both of which guide the update of the generator through its own loss. The parameters of the VGG16 feature extraction network are fixed and are not updated during training. Specifically, given a pretrained feature extraction network VGG16 $\phi(\cdot)$, let $\phi_i(\cdot)$ denote the ith output feature of the network for input image $\boldsymbol{x}$. The feature similarity loss represents the difference between two feature representations, which is shown in the following equation:

$$\mathcal{L}_2 = \|\phi_i(T(\boldsymbol{x}')) - \phi_i(\boldsymbol{x})\|_1. \tag{6}$$

In summary, the total loss function used to optimize the SMUNet is given by

$$\mathcal{L} = \mathcal{L}_1 + \mathcal{L}_2. \tag{7}$$

C. DO Strategy

In the RS scene classification task, the robustness of the network is closely related to the interclass distance. If the distance between two classes is overly close: a small perturbation is sufficient to misclassify the network, while a stronger perturbation is required to misclassify when the distance is relatively large. Notably, RS images exhibit a distinct characteristic of high interclass similarity, which often leads the network to extract similar high-dimensional features and result in smaller interclass distances. The introduction of the DO strategy aims to increase the distances between classes. By identifying classes with close interclass distances, which indicate network underfitting, this strategy strengthens the learning of such content. Consequently, it leads to a more concentrated distribution of intraclass features and a more dispersed distribution of interclass features. The DO strategy expands the feature distances between classes, thus improving the accuracy and adversarial robustness of the network. The proposed DO strategy has three stages: base AT, intensive training, and restorative training. The cross-entropy loss of DO strategy shown in (8) is employed to optimize the target classification network

$$\mathcal{H}(\theta, \boldsymbol{x}, y) = -\sum_{c=1}^{N} y_c \log(\hat{y}_c) \tag{8}$$

where $\hat{y}_c$ denotes the probability of class c for network prediction and N is the number of categories.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 19:39:23 UTC from IEEE Xplore. Restrictions apply.

First, base AT is conducted with clean examples $\boldsymbol{x}$, adversarial examples $\boldsymbol{x}'$, clean reconstructed examples $\boldsymbol{x}_{\text{rec}}$, and adversarial reconstructed examples $\boldsymbol{x}'_{\text{rec}}$ to give a more robust network M_{adv}, and the loss is denoted as

$$\mathcal{L}' = \mathcal{H}(\theta, \boldsymbol{x}, y) + \mathcal{H}\big(\theta, \boldsymbol{x}', y\big) + \mathcal{H}(\theta, \boldsymbol{x}_{\text{rec}}, y) + \mathcal{H}\big(\theta, \boldsymbol{x}'_{\text{rec}}, y\big). \tag{9}$$

Given a clean sample $\boldsymbol{x}$ and adversarial samples $\boldsymbol{x}'$ generated from attacks like FGSM, $\boldsymbol{x}_{\text{rec}}$ and $\boldsymbol{x}'_{\text{rec}}$, respectively, denote their reconstructed samples generated by SMUNet.

Interclass distance pairs refer to pairing each class with all other classes, and subsequently computing the distance between the two classes in the feature space to determine their distinctiveness. If the distance between classes is small, it indicates that these two classes have low distinctiveness with underfitting. In such cases, these classes need to be involved in creating underfitting subsets for further training. Specifically, the high-dimensional features of each class are obtained from the last convolutional layer of the network M_{adv}, with which the interclass distance pairs (C_i, C_j) are calculated and gathered into a distance pair dictionary $\text{Dis} = \{C_i - C_j : d\}$. The distance threshold shown in (10) is calculated from the mean and standard deviation of Dis, and interclass distance pairs with d below the threshold are listed as underfitting pairs, that is, the network has the insufficient fitting ability for those parts. The pair with the smallest distance is written as (C_0, C_1), and then the first underfitting subset is (D_0, D_1). The second smallest distance pair is (C_1, C_2), due to D_1 is present in the previous underfitting subsets; the second underfitting subset is (D_0, D_1, D_2). If the second smallest distance pair is (C_2, C_3), in which the classes never appear, the second underfitting subset is (D_0, D_1, D_2, D_3)

$$\text{Limit}_{\text{dis}} = \text{Dis}_{\text{mean}} - \alpha \cdot \text{Dis}_{\text{std}} \tag{10}$$

where α is a hyperparameter.

Based on the network M_{adv} and the underfitting subsets, that is, (D_0, D_1), (D_0, D_1, D_2), etc., the intensive training on specified categories is performed to enhance the learning for the underfitting content to get the optimized networks.

Finally, given that training for partial underfitting content may impact the classification performance of the network for other classes; thus, the network is fine-tuned on all kinds of data for restoration and is called restorative training. In the end, the best-performing network M_i with high adversarial robustness is selected. Different from other approaches generally devoted to modifying the structure of the networks, the proposed DO strategy can more fully exploit the training data to improve the adversarial robustness without changing their structure.

IV. Experiments

A. Data Descriptions

To verify the effectiveness of the proposed RDAT, two high-resolution RS datasets, including the UCM dataset [12] and the AID [55], are utilized for the experiments. To allow for a comprehensive evaluation with other algorithms, the training ratios are set according to the relevant references. For the UCM dataset, the training ratios are set to 50% and 80%. For the AID dataset, the training ratios are fixed at 20% and 50%, the training set is divided randomly. The input images are resized to 256 × 256 without any other data preprocessing or data augmentation.

1) UCM Dataset: The UCM dataset is proposed by Yang et al. [12] at UCM and is manually extracted from the U.S. Geological Survey (USGS) National Map Urban Area Imagery collection. The scenes in the dataset include 21 land-use classes such as agricultural, beach, forest, and intersection, and each class has 100 RGB images of 256 × 256 pixels with a spatial resolution of 0.3 m/pixel.

2) AID Dataset: AID dataset is a large-scale AID released by Wuhan University, by collecting sample images from Google Earth imagery. It contains 30 aerial scene types such as airport, bridge, desert, stadium, and so on, where each category includes 220–420 topographic images of 600 × 600 pixels. The images in the AID dataset are captured by different remote imaging sensors, so the resolution varies widely from 0.5 to 8 m/pixel, making the AID dataset more challenging for RS scene classification than the single-source images like the UCM dataset.

B. Experimental Settings

1) Targets Networks: ResNet152 [26], ResNeXt101 [28], and SCNet50 [43] are chosen as the target networks, and the structure of the networks is not changed during the experiments.

2) Attack Methods: Three typical attacks: FGSM [30], PGD [36], and CW [47] are selected to generate the adversarial examples. For FGSM, the perturbation parameter ε is set as 0.02. For the parameters of PGD, $\varepsilon = 4/255$, $\alpha = 0.05$, and the number of attack iterations is 20. PGD attack is considered the strongest first-order attack. For CW, $c = 0.1$ and $k = 20$, it is an optimization-based attack.

3) Experimental Design: The experiments in this article are mainly divided into five parts, the first part is adversarial attack, where adversarial examples are generated based on three attack methods, then the quantitative classification results and visualization examples are reported. The second part is the adversarial defense, where a series of experiments are conducted to test the defense ability of the networks under different attack methods on different datasets. The comparison methods include AT [36] and TRADES [39] proposed in computer vision, and PSGAN [41] for RS scene classification. Then the heat maps of the last convolution layer are visualized. The third part provides an analysis of the hyperparameter α in (10). The fourth part is the ablation experiment to prove the effectiveness of RA block and DO strategy in the proposed RDAT. Finally, the fifth part is a further discussion. On one hand, SwinT [56] (including tiny and small) has been introduced to evaluate the adaptability of RDAT. On the other hand, adversarial attack methods designed for RS scene classification, namely Mixup-Attack and its simple variant Mixcut-Attack [57], have been introduced to assess the comprehensive effectiveness of RDAT.

To quantitatively evaluate the defense methods, overall accuracy (OA, i.e., the ratio of correctly classified images

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 19:39:23 UTC from IEEE Xplore. Restrictions apply.

TABLE I
CLASSIFICATION ACCURACY WITH CLEAN AND VARIOUS ADVERSARIAL EXAMPLES

Methods		ResNet152	ResNeXt101	SCNet
UCM (50%)	clean	97.71	97.71	97.33
	FGSM	67.71	47.71	64.00
	PGD	36.19	21.33	28.66
	CW	60.00	36.67	48.95
UCM (80%)	clean	99.52	99.29	97.86
	FGSM	63.10	36.67	60.71
	PGD	30.24	25.00	46.90
	CW	58.57	31.90	60.00
AID (20%)	clean	91.04	93.54	92.03
	FGSM	13.68	32.74	44.08
	PGD	2.86	11.64	12.75
	CW	5.78	23.03	31.68
AID (50%)	clean	93.82	95.60	94.64
	FGSM	14.10	17.68	42.92
	PGD	1.88	3.84	8.30
	CW	5.20	1.43	28.72

to the number of images in the test set) is utilized as the evaluation metric, which is the most commonly used metric for RS scene classification. In (10), the value of α is set to 2.0. All experiments are implemented on NVIDIA GeForce GTX 3090 GPUs and PyTorch deep learning framework. The target networks are optimized using Adam [58] with a learning rate of 0.0001.

C. Adversarial Attack

In this section, the adversarial examples are first generated by three attacks, and then the quantitative classification results are reported on both the clean test sets and the adversarial test sets for three target networks. The experimental results are summarized in Table I.

First, it can be observed that the target networks have a relatively good performance on clean datasets, with the OA reaching over 90.00%. However, the networks perform unsatisfactorily when exposed to adversarial examples. Taking AID (20%) as an example, the OA of ResNeXt101 decreases from 93.54% on clean examples to 11.64% with PGD attack, where the OA even drops by 81.90%. The OA decreases by 30.00%–94.17% with different experimental datasets and target networks. This shows that it is not a coincidence that the adversarial examples will degrade network performance and prove the existence of network vulnerability. Moreover, with the set parameters, PGD shows the most robust attack performance, while FGSM is the weakest. This is attributed to the fact that PGD is a kind of iterative FGSM, which has stronger attack performance but takes more computation and time. Besides, it can also be observed that the effect of the attack is related to the size of the dataset, the larger the training data, the stronger the attack can be achieved. As shown in Table I, on different attacks and target networks, for UCM (50%) with relatively few training data, the OA decreases by 30.00%–76.38%, while for AID (50%) with relatively more training data, the OA decreases by 51.72%–94.17%.

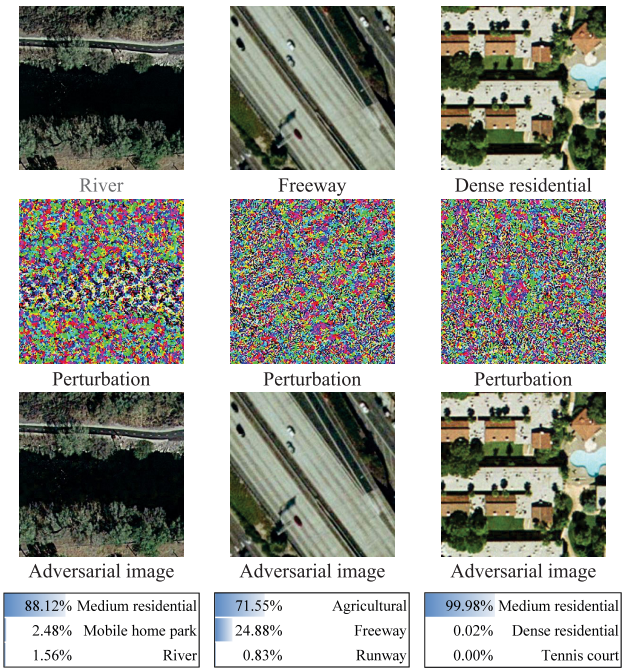


Fig. 4. Adversarial examples and prediction results on UCM (80%) with PGD attack. The bottom blue bars show the top three predicted categories and their probabilities.

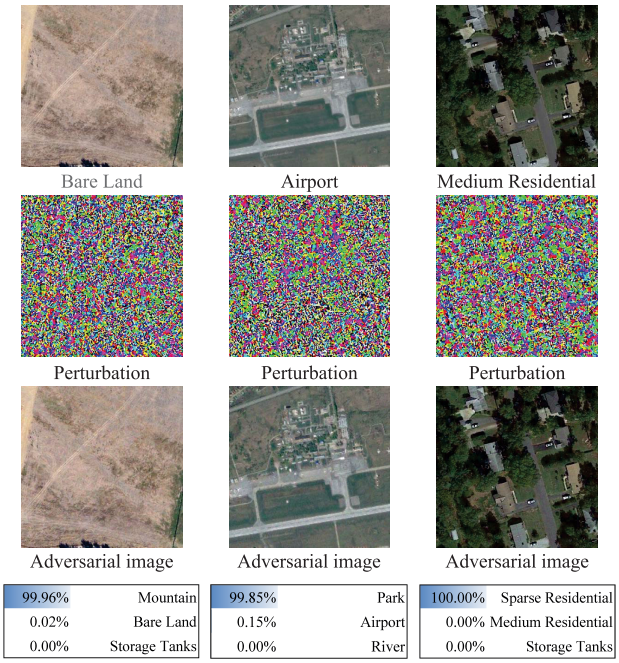


Fig. 5. Adversarial examples and prediction results on AID (50%) with PGD attack. The bottom blue bars show the top three predicted categories and their probabilities.

The examples of the adversarial examples and predicted results are shown in Figs. 4 and 5. In both figures, the perturbations shown in the second row are haphazard, and the difference between the clean examples in the first row and the adversarial examples in the third row is imperceptible to humans. However, these adversarial examples are enough to fool the networks and make them output the wrong classification with high confidence, which can be seen from the bottom blue bars.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 19:39:23 UTC from IEEE Xplore. Restrictions apply.

TABLE II
OVERALL ACCURACY (%) OF DIFFERENT METHODS ON UCM (50%)

UCM (50%)		ResNet152				ResNeXt101				SCNet			
train	test	AT	TRADES	PSGAN	RDAT	AT	TRADES	PSGAN	RDAT	AT	TRADES	PSGAN	RDAT
FGSM	FGSM	92.76	90.57	94.00	**96.48**	92.57	90.57	94.76	**95.43**	92.95	93.14	90.02	**97.14**
	PGD	94.86	91.05	96.55	**97.05**	96.00	95.05	96.74	**96.95**	94.38	94.19	93.16	**97.71**
	CW	93.14	90.86	94.21	**96.38**	94.29	93.14	94.88	**96.38**	92.48	92.76	90.44	**96.85**
PGD	FGSM	94.67	91.81	88.15	**95.52**	93.05	91.14	93.37	**94.67**	93.52	95.05	89.03	**95.33**
	PGD	96.48	92.19	95.87	**97.14**	95.14	96.29	96.44	**96.76**	95.24	96.00	94.53	**96.38**
	CW	94.76	91.52	93.37	**95.90**	93.81	94.00	95.29	**95.71**	93.24	95.05	90.06	**95.33**
CW	FGSM	92.38	89.52	88.53	**95.43**	90.67	89.05	93.20	**95.24**	92.76	95.14	89.36	**95.24**
	PGD	94.38	89.81	95.54	**96.29**	94.67	95.24	96.31	**97.14**	93.62	**95.81**	94.26	95.71
	CW	92.67	89.81	95.01	**96.10**	93.43	91.71	96.55	**96.86**	93.90	95.33	94.12	**95.81**

TABLE III
OVERALL ACCURACY (%) OF DIFFERENT METHODS ON UCM (80%)

UCM (80%)		ResNet152				ResNeXt101				SCNet			
train	test	AT	TRADES	PSGAN	RDAT	AT	TRADES	PSGAN	RDAT	AT	TRADES	PSGAN	RDAT
FGSM	FGSM	95.48	95.48	97.33	**97.62**	96.67	94.52	97.52	**98.10**	94.29	93.10	94.15	**97.14**
	PGD	96.43	95.24	98.24	**98.57**	97.38	97.62	98.29	**98.57**	95.71	94.76	95.33	**97.38**
	CW	95.71	95.71	97.33	**97.85**	97.38	95.71	97.69	**98.33**	94.76	93.81	94.14	**96.90**
PGD	FGSM	95.48	95.00	92.71	**96.67**	94.05	94.29	**96.86**	96.43	93.33	93.33	94.91	**97.38**
	PGD	98.10	95.71	97.28	**98.81**	96.67	97.62	98.19	**98.33**	96.67	95.48	95.99	**98.81**
	CW	95.48	95.00	96.61	**97.38**	95.00	95.71	**98.05**	97.62	94.29	94.29	95.76	**97.86**
CW	FGSM	94.29	91.67	92.95	**96.67**	95.00	93.57	96.53	**98.33**	95.48	91.90	95.10	**96.67**
	PGD	96.19	91.67	97.32	**97.86**	97.14	95.48	98.19	**98.57**	96.19	95.24	97.15	**98.10**
	CW	95.00	92.14	**98.15**	97.86	96.43	94.76	98.48	**98.57**	95.71	93.57	96.43	**97.62**

TABLE IV
OVERALL ACCURACY (%) OF DIFFERENT METHODS ON AID (20%)

AID (20%)		ResNet152				ResNeXt101				SCNet			
train	test	AT	TRADES	PSGAN	RDAT	AT	TRADES	PSGAN	RDAT	AT	TRADES	PSGAN	RDAT
FGSM	FGSM	77.73	80.44	83.78	**86.73**	85.30	82.73	86.54	**89.83**	86.46	87.13	81.98	**91.71**
	PGD	85.86	86.74	88.05	**89.44**	88.99	89.21	90.27	**90.85**	89.18	88.90	85.78	**92.18**
	CW	82.93	84.59	80.14	**88.85**	87.38	86.65	86.82	**90.73**	87.20	87.98	81.59	**91.11**
PGD	FGSM	58.74	74.23	75.31	**80.20**	84.71	82.39	84.85	**88.73**	84.09	87.51	77.67	**91.38**
	PGD	82.91	84.81	87.18	**89.51**	90.00	90.23	90.25	**91.76**	88.75	89.84	85.37	**92.43**
	CW	71.51	80.84	80.59	**87.23**	87.66	87.20	86.13	**90.77**	85.54	88.19	79.31	**91.38**
CW	FGSM	75.00	77.81	74.93	**85.64**	83.99	82.64	84.18	**88.91**	86.25	87.95	81.71	**90.46**
	PGD	85.65	85.26	86.40	**89.54**	88.39	88.90	89.64	**91.49**	87.86	89.99	88.37	**90.89**
	CW	82.23	83.04	87.32	**89.06**	87.30	86.78	89.47	**90.84**	86.84	88.94	87.18	**90.85**

D. Adversarial Defense

In this section, the existing defense methods AT [36], TRADES [39], and PSGAN [41] are utilized for comparison. Experiments are conducted on the UCM (with 50%, 80% training ratio) and AID (with 20%, 50% training ratio), where ResNet152, ResNeXt101, and SCNet50 are served as the target networks.

The experimental results are summarized in Tables II–V, where the highest OA is shown in bold. It can be observed that the proposed RDAT has a better performance against adversarial attacks compared to the other three algorithms. Moreover, for the UCM (with two training ratios on three target networks), as shown in Table I, the OA with adversarial examples on base networks is 21.33%–67.71%, which increases to 95.43%–98.81% with the proposed defense framework RDAT. For AID (with two training ratios on three target networks), the OA rises from 1.43%–44.08% to 86.73%–94.94%. This demonstrates that RDAT can effectively resist attacks and improve the robustness of the networks in the RS scene classification.

In practice, the network is likely to be exposed to many types of attacks, including attacks unknown to the network. An excellent robust network should perform well not only on known attacks, but also on other attacks. To evaluate the generalization of RDAT against other unknown attacks, training, and testing are chosen as different attacks. As can be observed in Tables II–V, RDAT performs well even when the test set uses other unknown attacks. Compared with the other three methods, RDAT has a stronger generalization ability (i.e., higher OA) for unknown adversarial examples. Besides,

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 19:39:23 UTC from IEEE Xplore. Restrictions apply.

TABLE V
OVERALL ACCURACY (%) OF DIFFERENT METHODS ON AID (50%)

AID (50%)		ResNet152				ResNeXt101				SCNet			
train	test	AT	TRADES	PSGAN	RDAT	AT	TRADES	PSGAN	RDAT	AT	TRADES	PSGAN	RDAT
FGSM	FGSM	88.76	87.56	91.34	**92.58**	91.06	89.06	91.86	**93.54**	91.16	91.50	90.53	**94.60**
	PGD	91.92	90.70	93.34	**93.44**	93.20	93.14	93.62	**94.06**	92.50	92.58	92.41	**94.68**
	CW	91.18	89.70	89.22	**93.44**	92.30	90.96	90.19	**93.68**	91.70	92.14	89.80	**94.36**
PGD	FGSM	74.34	79.74	86.99	**87.66**	85.80	83.86	89.97	**91.74**	91.48	88.82	89.40	**94.42**
	PGD	89.60	90.40	91.55	**93.00**	92.38	93.10	93.47	**94.66**	93.34	90.94	92.86	**94.94**
	CW	82.50	86.04	88.44	**90.98**	88.84	88.26	90.33	**93.28**	92.12	89.90	90.15	**94.44**
CW	FGSM	86.36	84.66	84.42	**90.22**	89.64	88.30	88.54	**92.74**	91.54	91.24	87.44	**94.36**
	PGD	91.78	89.52	91.73	**92.80**	92.68	93.00	92.62	**94.10**	92.80	92.52	90.52	**94.74**
	CW	90.48	87.88	92.12	**92.64**	91.30	90.96	92.76	**93.58**	92.46	91.92	88.35	**94.48**

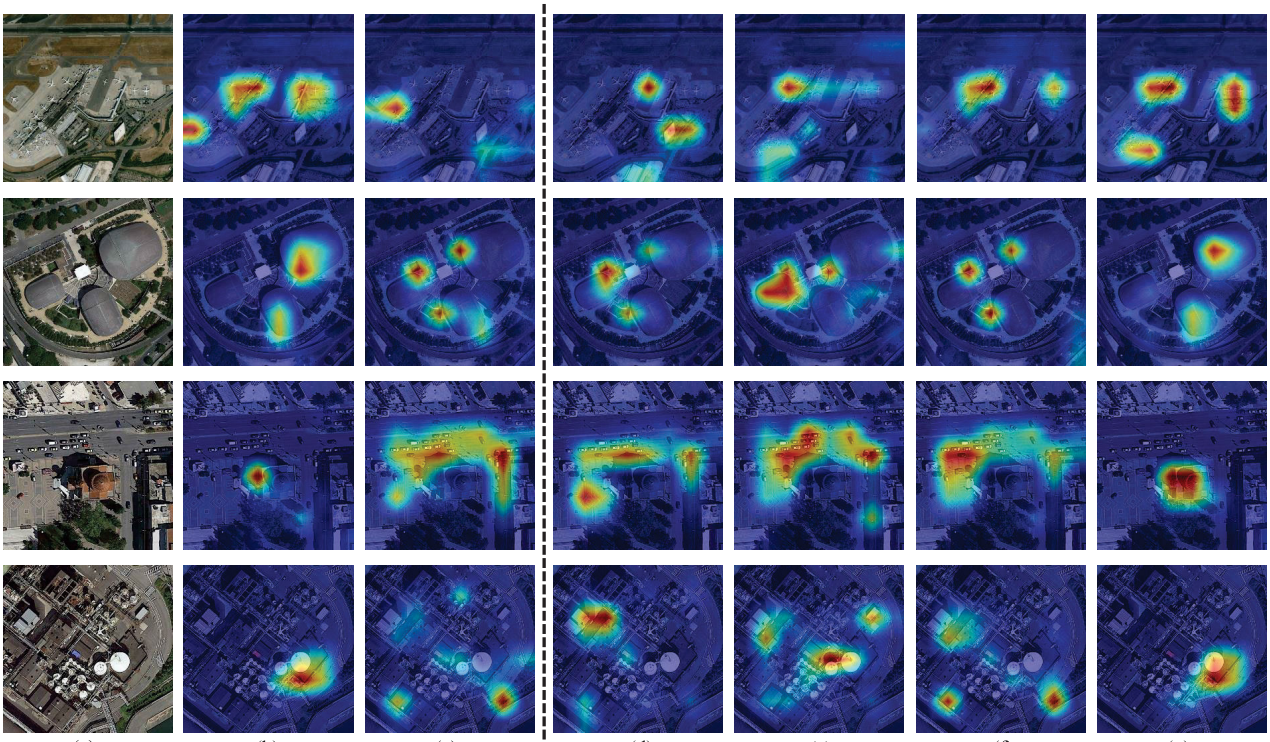


Fig. 6. Heat maps. (a) Clean image; (b) and (c) heat maps of the clean and attack image on the base network; (d)–(f) heat maps of the attack image with defense methods AT, TRADES, and PSGAN; and (g) heat map with RDAT.

Tables II–V also illustrate that RDAT can be applied to different target networks, and all of them achieve satisfactory defense performance, showing the strong applicability of the RDAT.

To further illustrate the effectiveness of RDAT for adversarial defense, the gradient-weighted class activation mapping (Grad-CAM) [59] is used to generate heat maps to visualize the regions of interest to the networks. As shown in Fig. 6, the experiment selects four images from the AID (20%) test set, where the target network is ResNet152, and the target layer is the last convolution layer of the target network.

First, it can be observed from Fig. 6(b) that when clean images are fed into the original base network, the network focuses precisely on salient objects, which means that discriminative features are extracted and excellent classification performance can be achieved. Then, the adversarial examples generated by FGSM are fed into the base network. From Fig. 6(c), it can be concluded that the network cannot effectively capture the saliency region of the targets in the adversarial examples, which reflects that the attack can disturb the feature extraction of the network, and makes the network fail to learn effective information. Finally, comparing the four defense algorithms in Fig. 6(d)–(g), it can be seen that the heat maps generated by AT, TRADES, and PSGAN still have a significant shift in the area of attention against adversarial examples and cannot fully remove the misinformation introduced by the attack. In contrast, it can be obtained from the heat maps generated by RDAT in Fig. 6(g) that RDAT enables the network to capture the core region of the scene, which is closest to the heat maps in Fig. 6(b). This confirms that RDAT well resists the attack and obtains a more robust network. The reasons for this can be attributed to the fact that, at the input data level, the RA block extracts the discriminative features and reduces the influence of the perturbations added in the

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 19:39:23 UTC from IEEE Xplore. Restrictions apply.

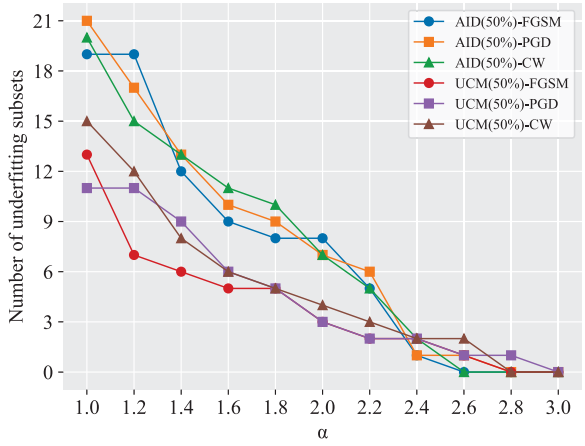


Fig. 7. Number of underfitting subsets with different values of α on UCM (50%) and AID (50%) with ResNet152.

attack images, and at the model level, DO strategy focuses on the interclass feature distance of RS images and enhances the learning of underfitting content to obtain a more robust network.

E. Parameter Analysis

In statistics, points that fall below two times the standard deviation from the mean are considered outliers. These points deviate significantly from the mean of the dataset and lie far outside the distribution range of the data, requiring special attention. Based on this consideration, the parameter α in (10) is set to 2.0. If the feature distance between pairs of classes is less than the threshold defined in (10), it indicates they are significantly closer and the network suffers from underfitting. These underfitting class pairs are utilized in constructing the underfitting subsets.

To further investigate the impact of different values of α on the number of underfitting subsets, experiments are conducted with α taking values in the range [1.0, 1.2, 1.4, 1.6, 2.0, 2.2, 2.4, 2.6, 2.8, 3.0] on UCM (50%) and AID (50%) with ResNet152. As observed in Fig. 7, when the value of α is too small, there is an excessive number of underfitting subsets, resulting in high computational and time costs. Conversely, when the value of α is too large, too few subsets are included, and the underfitting content is not sufficiently included. Considering principles from statistics, as well as the performance and computational costs, it is concluded that setting the value of α to 2.0 strikes an appropriate balance. This configuration ensures a moderate number of underfitting subsets across different datasets and attacks, consequently enhancing the learning of underfitting content and improving the adversarial robustness of networks.

F. Ablation Experiments

To better prove the effectiveness of the blocks in the RDAT defense framework, ablation experiments are implemented on UCM (50%) and AID (50%) (with three attacks). The RA block and DO strategy are studied separately for their effects on defense effectiveness.

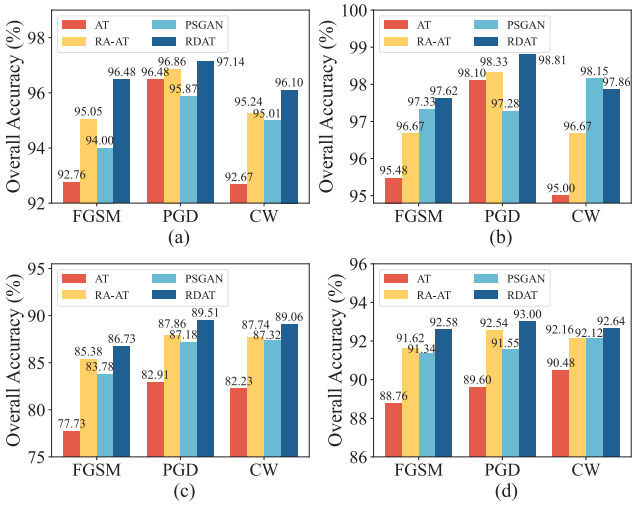


Fig. 8. Ablation experiments of the proposed RDAT on ResNet152. (a) UCM (50%). (b) UCM (80%). (c) AID (20%). (d) AID (50%).

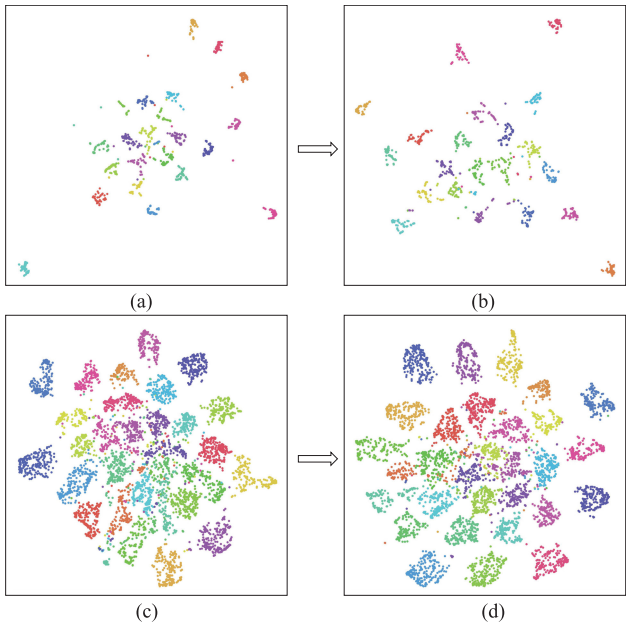


Fig. 9. Two-dimensional feature visualization, where different colors represent different classes of images. (a) and (b) Represent the feature visualization on the UCM (50%) with RA-AT and RDAT, respectively. (c) and (d) Represent the feature visualization on the AID (50%) with RA-AT and RDAT, respectively.

The experimental results are shown in Fig. 8(a)–(d), where "AT" refers to AT by adversarial examples, "RA-AT" represents AT with RA block, "PSGAN" is the latest defense algorithm in RS scene classification, and "RDAT" employs RA block and DO strategy. ResNet152 was chosen to do the ablation experiment.

1) Removing RA Block: The SMUNet is used to map the input adversarial examples to clean examples, generating reconstructed images that are used for AT along with the adversarial examples. Comparing the OA of "RA-AT" and "AT" as in Fig. 8(a)–(d), it can be observed that there is a significant decrease in OA without reconstructed images under various attacks and different datasets from "RA-AT" to "AT." Experimental results demonstrate that RA block can improve defense performance, which is mainly due to the fact that

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 19:39:23 UTC from IEEE Xplore. Restrictions apply.

TABLE VI
COMPARISON OF CONVOLUTION-BASED METHODS AND TRANSFORMER-BASED METHODS

Datasets	Networks	FGSM	PGD	CW
UCM (50%)	ResNet152	95.52	97.14	95.90
	ResNeXt101	94.67	96.76	95.71
	SCNet	95.33	96.38	95.33
	Swin-T	95.62	97.43	96.29
	Swin-S	**95.71**	**97.81**	**97.14**
AID (50%)	ResNet152	87.66	93.00	90.98
	ResNeXt101	91.74	94.66	93.28
	SCNet	**94.42**	94.94	**94.44**
	Swin-T	91.70	94.96	93.54
	Swin-S	91.90	**95.06**	93.66

TABLE VII
OVERALL ACCURACY (%) OF DIFFERENT DEFENSE METHODS AGAINST ATTACKS DESIGNED FOR RS SCENE CLASSIFICATION

Datasets	Attacks	AT	TRADES	PSGAN	RDAT
UCM (50%)	Mixup	95.43	91.90	94.76	**96.76**
	Mixcut	95.42	91.81	94.57	**96.76**
AID (50%)	Mixup	90.50	91.20	89.50	**93.14**
	Mixcut	90.86	91.42	89.90	**93.28**

SMUNet extracts rich global and multiscale local features and largely eliminates the disturbance of adversarial perturbations.

2) Removing DO Strategy: RS images have high interclass similarity, and it should be noted that the feature distance between classes is related to the robustness of the networks, thus DO strategy is used to enhance the learning of underfitting content and improves robustness from a distance perspective. Comparing the OA of "RA-AT" and "RDAT" in Fig. 8(a)–(d), it can be seen that the OA of RA-AT is lower than RDAT, that is, the RDAT further improves the performance in most of the experimental settings, due to the increase in interclass distance and the enhancement of the learning of underfitting content.

Fig. 9 shows the variation of interclass distances. Specifically, it shows the results of embedding high-dimensional features into 2-D space, where the colored points represent examples from different classes. Fig. 9(a) and (c) shown the 2-D features of adversarial examples with RA-AT, while Fig. 9(b) and (d) displays the features with RDAT. Meanwhile, from Fig. 9(a) and (b), it shows the change in feature distance after the introduction of DO strategy on UCM (50%), and Fig. 9(c) and (d) shows the corresponding result on AID (50%). It is easy to find that compared with RA-AT, RDAT has more concentrated clustering in the same class and more scattered examples in different classes, that is, the feature distance is farther. The same phenomenon is found in both UCM (50%) and AID (50%), indicating that the DO strategy can provide a more effective feature representation and improve the adversarial robustness of the networks.

Furthermore, to vividly demonstrate the superior performance of RDAT, a quantitative comparison is conducted with the latest defense algorithm in RS scene classification, PSGAN. As shown in Fig. 8, it can be observed that RDAT outperforms PSGAN in the majority of cases. Notably, when only the RA block (i.e., RA-AT) is introduced, it achieves better defense performance, even outperforming PSGAN. This outcome further validates the effectiveness of RDAT.

G. Discussion

1) Extension to Transformer-Based Methods: In addition to convolution-based methods (such as ResNet152), RDAT can also be effectively combined with transformer-based methods. SwinT (including tiny and small) has been incorporated for further analysis. As shown in Table VI, experiments are conducted on UCM (50%) and AID (50%) with PGD attack to evaluate the defense performance of both convolution-based and transformer-based methods. It can be observed that RDAT exhibits favorable adaptability when applied to transformer-based methods. Furthermore, it is worth noting that transformer-based methods outperform the majority of their convolution-based methods in the defense framework RDAT.

2) Testing on Scene Classification Attacks: To comprehensively assess the effectiveness of RDAT, adversarial attacks designed for RS scene classification, Mixup-Attack, and its simple variant Mixcut-Attack, are introduced. They are black-box attacks, and the surrogate network in the experiments is ResNet18. On UCM (50%) and AID (50%), based on PGD attacks, robust networks are trained on ResNet152 with defense algorithms AT, TRADES, PSGAN, and RDAT, respectively for defense testing. As shown in Table VII, it is evident that all these algorithms demonstrate defense capability against Mixup-Attack and Mixcut-Attack to some extent. Notably, RDAT yields the best results.

V. CONCLUSION

In this article, an effective defense framework RDAT is proposed to defend attacks in RS scene classification. On the one hand, the RA block is used to perform data preprocessing aiming to eliminate perturbations in the adversarial examples. Specifically, SMUNet is constructed to extract rich discriminative features with SM block. On the other hand, the DO strategy is used to enhance the learning of underfitting content and increase the interclass distance to improve the adversarial robustness. Extensive experiments on the benchmark datasets demonstrate the effectiveness and advancement of RDAT.

In future work, SMUNet for image reconstruction is planned to be optimized to obtain higher-quality reconstructed examples. Moreover, it is considered to be combined with AT and its enhanced form to improve the adversarial robustness of the networks and defend against more advanced attacks.

REFERENCES

[1] J. Chen et al., "Contextual information-preserved architecture learning for remote-sensing scene classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5602614.
[2] S. Mei et al., "Lightweight multiresolution feature fusion network for spectral super-resolution," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5501414.
[3] S. Wang, D. Hou, and H. Xing, "A self-supervised-driven open-set unsupervised domain adaptation method for optical remote sensing image scene classification and retrieval," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5605515.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 19:39:23 UTC from IEEE Xplore. Restrictions apply.

[4] J. Li, Y. Liu, R. Song, Y. Li, K. Han, and Q. Du, "Sal^{2}RN: A spatial–spectral salient reinforcement network for hyperspectral and LiDAR data fusion classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5500114.

[5] X.-Y. Tong et al., "Land-cover classification with high-resolution remote sensing images using transferable deep models," *Remote Sens. Environ.*, vol. 237, Feb. 2020, Art. no. 111322.

[6] Z. Y. Lv, W. Shi, X. Zhang, and J. A. Benediktsson, "Landslide inventory mapping from bitemporal high-resolution remote sensing images using change detection and multiscale segmentation," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 11, no. 5, pp. 1520–1532, May 2018.

[7] Q. Zhu et al., "Knowledge-guided land pattern depiction for urban land use mapping: A case study of Chinese cities," *Remote Sens. Environ.*, vol. 272, Apr. 2022, Art. no. 112916.

[8] G. Cheng, X. Xie, J. Han, L. Guo, and G.-S. Xia, "Remote sensing image scene classification meets deep learning: Challenges, methods, benchmarks, and opportunities," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 13, pp. 3735–3756, 2020.

[9] J. Zhang, M. Marszałek, S. Lazebnik, and C. Schmid, "Local features and kernels for classification of texture and object categories: A comprehensive study," *Int. J. Comput. Vis.*, vol. 73, no. 2, pp. 213–238, Jun. 2007.

[10] G. Cheng, J. Han, P. Zhou, and L. Guo, "Multi-class geospatial object detection and geographic image classification based on collection of part detectors," *ISPRS J. Photogramm. Remote Sens.*, vol. 98, pp. 119–132, Dec. 2014.

[11] E. Aptoula, "Remote sensing image retrieval with global morphological texture descriptors," *IEEE Trans. Geosci. Remote Sens.*, vol. 52, no. 5, pp. 3023–3034, May 2014.

[12] Y. Yang and S. Newsam, "Bag-of-visual-words and spatial extensions for land-use classification," in *Proc. 18th SIGSPATIAL Int. Conf. Adv. Geographic Inf. Syst.*, Nov. 2010, pp. 270–279.

[13] F. Perronnin and C. Dance, "Fisher kernels on visual vocabularies for image categorization," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, Jun. 2007, pp. 1–8.

[14] X. Chen, Z. Han, Y. Li, M. Ma, S. Mei, and W. Cheng, "Attention-aware deep feature embedding for remote sensing image scene classification," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 16, pp. 1171–1184, 2023.

[15] X. Zheng, Y. Yuan, and X. Lu, "A deep scene representation for aerial scene classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 7, pp. 4799–4809, Jul. 2019.

[16] S. Mei, J. Lian, X. Wang, Y. Su, M. Ma, and L.-P. Chau, "A comprehensive study on the robustness of image classification and object detection in remote sensing: Surveying and benchmarking," 2023, *arXiv:2306.12111*.

[17] W. Li, J. Wang, Y. Gao, M. Zhang, R. Tao, and B. Zhang, "Graph-feature-enhanced selective assignment network for hyperspectral and multispectral data classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5526914.

[18] J. Wang, F. Gao, J. Dong, and Q. Du, "Adaptive DropBlock-enhanced generative adversarial networks for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 6, pp. 5040–5053, Jun. 2021.

[19] X. Hu et al., "SPNet: Spectral patching end-to-end classification network for UAV-borne hyperspectral imagery with high spatial and spectral resolutions," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5503417.

[20] J. Wang, W. Li, Y. Gao, M. Zhang, R. Tao, and Q. Du, "Hyperspectral and SAR image classification via multiscale interactive fusion network," *IEEE Trans. Neural Netw. Learn. Syst.*, early access, May 11, 2022, doi: 10.1109/TNNLS.2022.3171572.

[21] J. Wang, F. Gao, J. Dong, S. Zhang, and Q. Du, "Change detection from synthetic aperture radar images via graph-based knowledge supplement network," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 15, pp. 1823–1836, 2022.

[22] J. Wang, F. Gao, J. Dong, Q. Du, and H.-C. Li, "Change detection from synthetic aperture radar images via dual path denoising network," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 15, pp. 2667–2680, 2022.

[23] N. Kussul, M. Lavreniuk, S. Skakun, and A. Shelestov, "Deep learning classification of land cover and crop types using remote sensing data," *IEEE Geosci. Remote Sens. Lett.*, vol. 14, no. 5, pp. 778–782, May 2017.

[24] G. J. Scott, M. R. England, W. A. Starms, R. A. Marcum, and C. H. Davis, "Training deep convolutional neural networks for land–cover classification of high-resolution imagery," *IEEE Geosci. Remote Sens. Lett.*, vol. 14, no. 4, pp. 549–553, Apr. 2017.

[25] C. Szegedy et al., "Going deeper with convolutions," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2015, pp. 1–9.

[26] K. He, X. Zhang, S. Ren, and J. Sun, "Deep residual learning for image recognition," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2016, pp. 770–778.

[27] G. Huang, Z. Liu, L. Van Der Maaten, and K. Q. Weinberger, "Densely connected convolutional networks," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jul. 2017, pp. 2261–2269.

[28] S. Xie, R. Girshick, P. Dollár, Z. Tu, and K. He, "Aggregated residual transformations for deep neural networks," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jul. 2017, pp. 1492–1500.

[29] M. Tan and Q. Le, "EfficientNet: Rethinking model scaling for convolutional neural networks," in *Proc. 36th Int. Conf. Mach. Learn.*, 2019, pp. 6105–6114.

[30] I. J. Goodfellow, J. Shlens, and C. Szegedy, "Explaining and harnessing adversarial examples," 2014, *arXiv:1412.6572*.

[31] A. Kurakin, I. Goodfellow, and S. Bengio, "Adversarial examples in the physical world," 2016, *arXiv:1607.02533*.

[32] N. Papernot, P. McDaniel, and I. Goodfellow, "Transferability in machine learning: From phenomena to black-box attacks using adversarial samples," 2016, *arXiv:1605.07277*.

[33] G. Cheng, J. Han, and X. Lu, "Remote sensing image scene classification: Benchmark and state of the art," *Proc. IEEE*, vol. 105, no. 10, pp. 1865–1883, Oct. 2017.

[34] W. Wang, Y. Chen, and P. Ghamisi, "Transferring CNN with adaptive learning for remote sensing scene classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5533918.

[35] Y. Xu, B. Du, and L. Zhang, "Assessing the threat of adversarial examples on deep neural networks for remote sensing scene classification: Attacks and defenses," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 2, pp. 1604–1617, Feb. 2021.

[36] A. Madry, A. Makelov, L. Schmidt, D. Tsipras, and A. Vladu, "Towards deep learning models resistant to adversarial attacks," 2017, *arXiv:1706.06083*.

[37] N. Papernot, P. McDaniel, X. Wu, S. Jha, and A. Swami, "Distillation as a defense to adversarial perturbations against deep neural networks," in *Proc. IEEE Symp. Secur. Privacy (SP)*, May 2016, pp. 582–597.

[38] P. Samangouei, M. Kabkab, and R. Chellappa, "Defense-GAN: Protecting classifiers against adversarial attacks using generative models," 2018, *arXiv:1805.06605*.

[39] H. Zhang, Y. Yu, J. Jiao, E. Xing, L. El Ghaoui, and M. Jordan, "Theoretically principled trade-off between robustness and accuracy," in *Proc. 36th Int. Conf. Mach. Learn.*, 2019, pp. 7472–7482.

[40] L. Chen, J. Xiao, P. Zou, and H. Li, "Lie to me: A soft threshold defense method for adversarial examples of remote sensing images," *IEEE Geosci. Remote Sens. Lett.*, vol. 19, 2022, Art. no. 8016905.

[41] G. Cheng, X. Sun, K. Li, L. Guo, and J. Han, "Perturbation-seeking generative adversarial networks: A defense framework for remote sensing image scene classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5605111.

[42] T. Gao, Z. Liu, J. Zhang, G. Wu, and T. Chen, "A task-balanced multiscale adaptive fusion network for object detection in remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5613515.

[43] J.-J. Liu, Q. Hou, M.-M. Cheng, C. Wang, and J. Feng, "Improving convolutional networks with self-calibrated convolutions," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, 2020, pp. 10096–10105.

[44] C. Szegedy et al., "Intriguing properties of neural networks," 2013, *arXiv:1312.6199*.

[45] J. Long, E. Shelhamer, and T. Darrell, "Fully convolutional networks for semantic segmentation," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, 2015, pp. 3431–3440.

[46] N. Papernot, P. McDaniel, I. Goodfellow, S. Jha, Z. B. Celik, and A. Swami, "Practical black-box attacks against machine learning," in *Proc. ACM Asia Conf. Comput. Commun. Secur.*, Apr. 2017, pp. 506–519.

[47] N. Carlini and D. Wagner, "Towards evaluating the robustness of neural networks," in *Proc. IEEE Symp. Secur. Privacy (SP)*, May 2017, pp. 39–57.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 19:39:23 UTC from IEEE Xplore. Restrictions apply.

[48] N. Akhtar and A. Mian, "Threat of adversarial attacks on deep learning in computer vision: A survey," *IEEE Access*, vol. 6, pp. 14410–14430, 2018.
[49] S. Y. Khamaiseh, D. Bagagem, A. Al-Alaj, M. Mancino, and H. W. Alomari, "Adversarial deep learning: A survey on adversarial attacks and defense mechanisms on image classification," *IEEE Access*, vol. 10, pp. 102266–102291, 2022.
[50] G. Hinton, O. Vinyals, and J. Dean, "Distilling the knowledge in a neural network," 2015, *arXiv:1503.02531*.
[51] A. Ross and F. Doshi-Velez, "Improving the adversarial robustness and interpretability of deep neural networks by regularizing their input gradients," in *Proc. AAAI Conf. Artif. Intell.*, vol. 32, no. 1, 2018, pp. 1–10.
[52] G. Karolina Dziugaite, Z. Ghahramani, and D. M. Roy, "A study of the effect of JPG compression on adversarial images," 2016, *arXiv:1608.00853*.
[53] J. Lu, T. Issaranon, and D. Forsyth, "SafetyNet: Detecting and rejecting adversarial examples robustly," in *Proc. IEEE Int. Conf. Comput. Vis. (ICCV)*, Oct. 2017, pp. 446–454.
[54] W. Xu, D. Evans, and Y. Qi, "Feature squeezing: Detecting adversarial examples in deep neural networks," 2017, *arXiv:1704.01155*.
[55] G.-S. Xia et al., "AID: A benchmark data set for performance evaluation of aerial scene classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 55, no. 7, pp. 3965–3981, Jul. 2017.
[56] Z. Liu et al., "Swin transformer: Hierarchical vision transformer using shifted windows," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2021, pp. 10012–10022.
[57] Y. Xu and P. Ghamisi, "Universal adversarial examples in remote sensing: Methodology and benchmark," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5619815.
[58] D. P. Kingma and J. Ba, "Adam: A method for stochastic optimization," 2014, *arXiv:1412.6980*.
[59] R. R. Selvaraju, M. Cogswell, A. Das, R. Vedantam, D. Parikh, and D. Batra, "Grad-CAM: Visual explanations from deep networks via gradient-based localization," in *Proc. IEEE Int. Conf. Comput. Vis. (ICCV)*, Oct. 2017, pp. 618–626.



Yuru Su received the B.Eng. degree in electronics and information engineering from Northwestern Polytechnical University, Xi'an, China, in 2021, where she is currently pursuing the D.Eng. degree in new generation of electronic information technology with the School of Electronics and Information.

Her main research interests include remote sensing and image processing.



Ge Zhang (Graduate Student Member, IEEE) received the B.S. degree in electronics and information engineering and the Ph.D. degree in communication and information systems from Northwestern Polytechnical University, Xi'an, China, in 2018 and 2023, respectively.

Her research interests include hyperspectral image unmixing and signal processing.



Shaohui Mei (Senior Member, IEEE) received the B.S. degree in electronics and information engineering and the Ph.D. degree in signal and information processing from Northwestern Polytechnical University, Xi'an, China, in 2005 and 2011, respectively.

He was a Visiting Student with the University of Sydney, Camperdown, NSW, Australia, from October 2007 to October 2008. He is currently a Professor with the School of Electronics and Information, Northwestern Polytechnical University. His research interests include hyperspectral remote sensing image processing and applications, intelligent signal and information acquisition and processing, video processing, and pattern recognition.

Dr. Mei received the First Prize of Natural Science Award of Shaanxi Province in 2022, the Excellent Doctoral Dissertation Award of Shaanxi Province in 2014, the Best Paper Award of the IEEE International Symposium on Intelligent Signal Processing and Communication Systems (ISPACS) in 2017, and the Best Reviewer of IEEE JOURNAL OF SELECTED TOPICS IN APPLIED EARTH OBSERVATIONS AND REMOTE SENSING (JSTARS) in 2019, and IEEE TRANSACTIONS ON GEOSCIENCE AND REMOTE SENSING (TGRS) in 2022. He is serving as an Associate Editor for IEEE TGRS and IEEE JSTARS, a Guest Editor for *Remote Sensing*, and a reviewer for more than 30 international famous academic journals.



Jiawei Lian (Graduate Student Member, IEEE) received the B.Eng. degree in automation from Jiangxi University of Science and Technology, Ganzhou, China, in 2019, and the M.Eng. degree in control engineering from Northwestern Polytechnical University, Xi'an, China, in 2022. He is currently pursuing the Ph.D. degree in information and communication engineering with the School of Electronics and Information, Northwestern Polytechnical University, Xi'an, China.

His research interests include trustworthy machine learning and remote sensing.



Ye Wang (Graduate Student Member, IEEE) received the B.S. degree in automation from the Xi'an University of Science and Technology, Xi'an, China, in 2017, and the M.S. degree in biomedical engineering from Northwestern Polytechnical University, Xi'an, in 2020, where she is currently pursuing the Ph.D. degree in information and communication engineering with the School of Electronics and Information.

Her research interests include remote sensing, hyperspectral object tracking, image processing, and deep learning.



Shuai Wan (Member, IEEE) received the B.E. degree in telecommunication engineering and the M.E. degree in communication and information systems from Xidian University, Xi'an, China, in 2001 and 2004, respectively, and the Ph.D. degree in electronics engineering from the Queen Mary University of London, London, U.K., in 2007.

She is currently a Professor with Northwestern Polytechnical University, Xi'an, and an Adjunct Professor with RMIT University, Melbourne VIC, Australia. Her research interests include point cloud compression, versatile video coding, neural networks-based video coding, and multimedia processing.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 19:39:23 UTC from IEEE Xplore. Restrictions apply.